CENTRAL MANUFACTURING TECHNOLOGY INSTITUTE

C – Smart Manufacturing Cell

User Guide

Troubleshooting-

Azure Storage Monitor

PROJECT: SPV Data Flow

Azure Storage Monitor

The Azure Storage Monitor module provides real-time visibility into Azure Blob Storage connectivity, upload scheduler status, operation logs, and source path configuration. It contains four tabs: Schedule Status, Connectivity, Logs, and Path Config.

Tab	Purpose
Schedule Status	View the Azure Upload Scheduler status, last upload time, next scheduled upload, and last execution result for each machine type (EL, IVC, RFID).
Connectivity	Verify live connection status of Azure Blob Storage and PostgreSQL database. View container, account, endpoint, and schema details.
Logs	View real-time Azure upload operation logs including HTTP request/response headers and upload status codes.
Path Config	Manage source folder paths used by the upload service. Add, edit, activate, or deactivate paths per machine type.

Schedule Status – Last Execution Failed

If the Schedule Status tab shows Last Execution Status as "failed" for a machine type:

Step 1: Identify the affected machine type

→ Click through the ALL, EL, IVC, and RFID filter tabs to identify which machine type has failed. Note the Last Upload time and Next Scheduled Upload to understand when the failure occurred.

Step 2: Check Azure Blob Storage connectivity

→ Navigate to the Connectivity tab. Confirm Azure Blob Storage shows Connected (green dot). If it shows Disconnected, resolve the connection issue first.

Step 3: Inspect the Logs tab for error details

→ Navigate to the Logs tab and review the most recent upload operation entries. Look for HTTP error status codes (4xx or 5xx) or authentication errors in the response headers. This will identify the exact failure reason.

Step 4: Verify the source path is active

→ Navigate to the Path Config tab. Confirm that at least one path for the affected machine type has Status: Active. If all paths are Inactive, the scheduler has no source to upload from and will fail every cycle.

Step 5: Trigger a manual upload to re-test

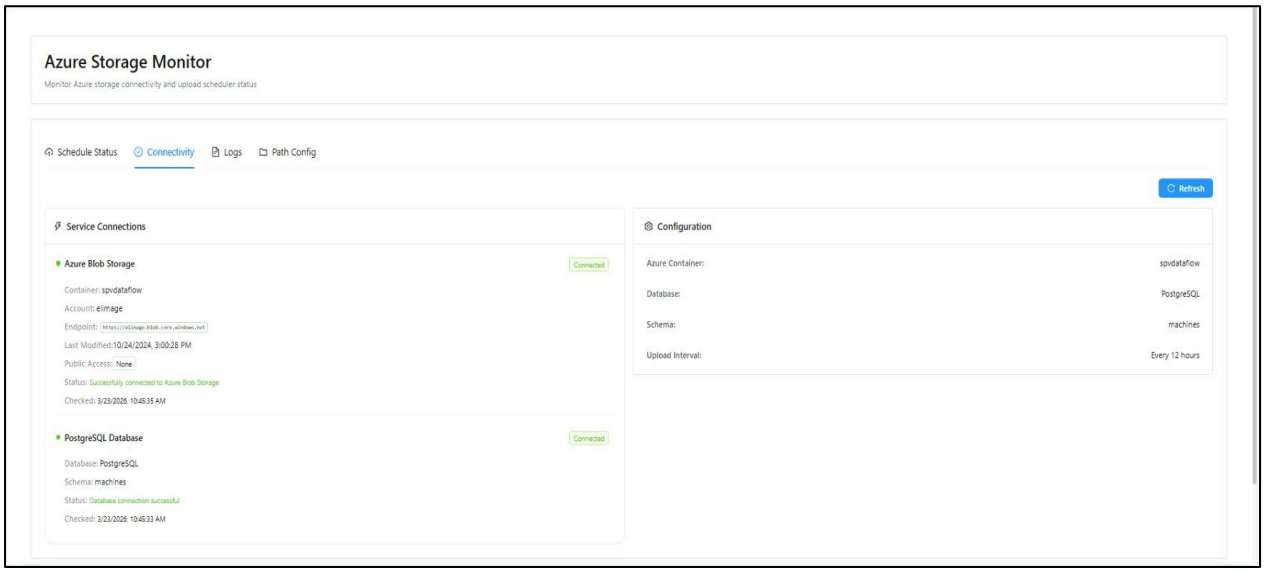
→ After resolving the identified issue, click on trigger button to manually trigger an upload without waiting for the next scheduled cycle.

Click Refresh on the Schedule Status tab to confirm the Last Execution Status changes to success.

NOTE: The scheduler runs automatically every 12 hours. A single failed execution does not stop future cycles. However, if the root cause is not fixed, every subsequent scheduled run will also fail.

Connectivity Tab – Understanding Service Status

The Connectivity tab shows live connection health for both Azure Blob Storage and PostgreSQL database. Use this tab as the first diagnostic step whenever uploads fail.



Key fields to verify on this tab:

Service	Field	What to Check
Azure Blob Storage	Status	Must show 'Successfully connected to Azure Blob Storage' in green.
Azure Blob Storage	Container	Confirm container name is spvdataflow (matches application config).
Azure Blob Storage	Account	Confirm account name is elimage (matches connection string).
Azure Blob Storage	Endpoint	Confirm endpoint URL matches the configured storage account.

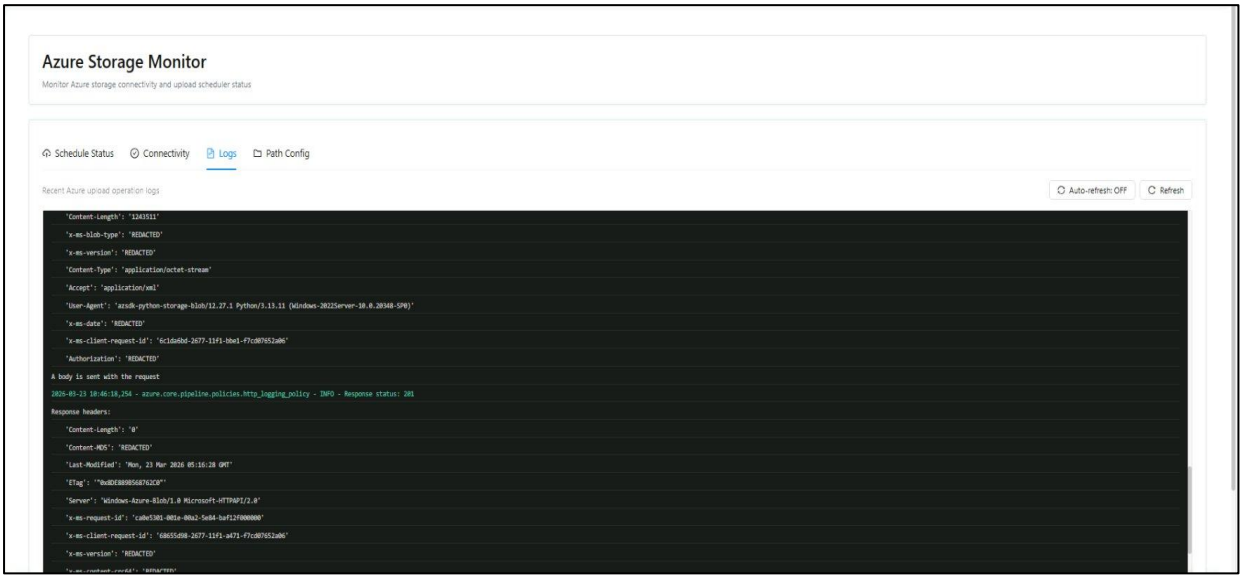
PostgreSQL Database	Status	Must show ‘Database connection successful’ in green.
PostgreSQL Database	Database	Confirm database name is PostgreSQL.
PostgreSQL Database	Schema	Confirm schema is machines.
Configuration	Upload Interval	Confirm Every 12 hours is shown – matches scheduler configuration.

Azure Blob Storage shows Disconnected

→ Verify the connection string is correct and the account key has not been rotated. Check outbound HTTPS (port 443) to *.blob.core.windows.net is not blocked. If on a private tunnel, confirm the private endpoint URL is configured.

Logs Tab – Reading Upload Operation Logs

The Logs tab shows raw Azure SDK upload operation logs. These are the most precise source of failure information when connectivity appears healthy but uploads still fail.



How to read the logs:

- Green/cyan lines prefixed with a timestamp are INFO-level log entries from the Azure SDK pipeline.
- Response status: 201 indicates a successful blob upload (HTTP 201 Created).
- Response status: 4xx indicates an authentication or permissions error.
- Response status: 5xx indicates an Azure-side server error.
- REDACTED fields (Authorization, x-ms-date, x-ms-version) are security headers – their presence confirms the SDK is forming requests correctly.

Step 1: Identify the HTTP response status code

→ Scroll to the most recent upload operation and find the line: Response status: <code>. A 201 confirms success. Any other code indicates a failure – note it and refer to the table below.

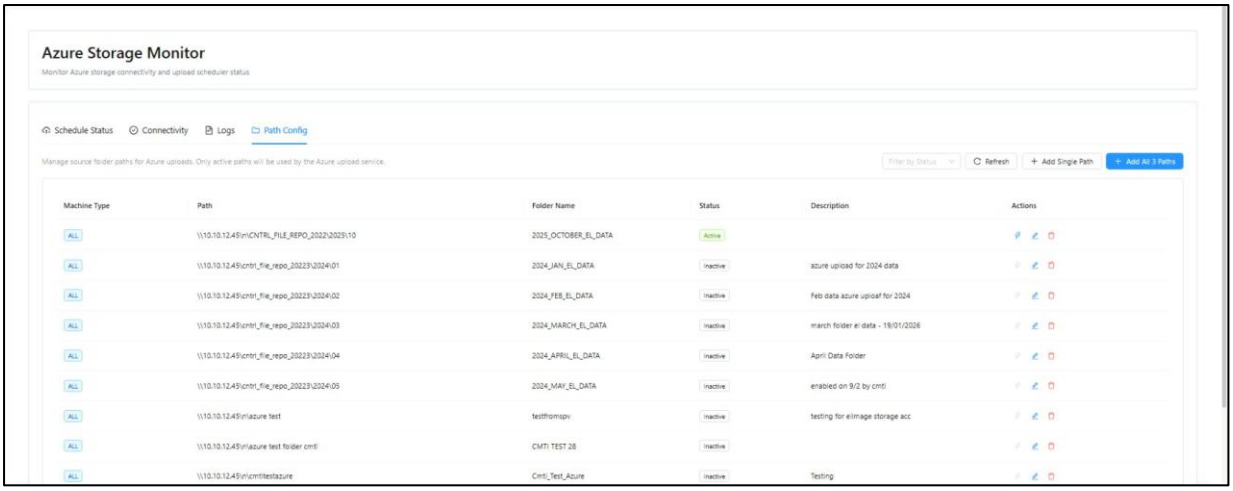
HTTP Code	Meaning	Action
201 Created	Blob uploaded successfully.	No action needed. Upload succeeded.
401 Unauthorized	Account key is invalid or expired.	Regenerate the storage account key and update the connection string.
403 Forbidden	Firewall rule or RBAC blocking access.	Check Azure Storage firewall rules; verify network access.
404 Not Found	Container or blob path does not exist.	Confirm container name (spvdataflow) exists in the storage account.
409 Conflict	Blob already exists and overwrite not allowed.	Expected in some cases – check if the upload still recorded as failed.
5xx Server Error	Azure-side issue.	Retry manually. If persistent, check Azure Service Health portal.

Step 2: Check User-Agent and SDK version

→ The User-Agent header in the logs (e.g. azsdk-python-storage-blob/12.27.1 Python/3.13.11) identifies the Azure SDK version in use. If logs stop appearing entirely, the SDK may have lost its connection to Azure – restart the backend container.

Path Config – Managing Source Upload Paths

The Path Config tab manages the source folder paths that the upload service scans for files. Only paths with Status: Active are used during scheduled or manual uploads.



Step 1: Adding a new path

→ Click + Add Single Path to register a new source folder. Fill in: Machine Type (ALL, EL, IVC, RFID), Path (UNC network path, e.g. \\10.10.12.45\n\folder), Folder Name

(display label), and Description. Set Status to Active to include it in uploads immediately.

Step 2: Path shows Inactive but should be uploading

→ Click the Edit (pencil) icon for the path and confirm the path is toggled Active. Also verify the UNC path is accessible from the server – network shares must be mounted and the server must have read access to \\10.10.12.45.

Step 4: Network share path is inaccessible

→ If the upload service cannot read from a UNC path (e.g. \\10.10.12.45\n\CNTRL_FILE_REPO_2022), confirm the network share is online.

NOTE: Source paths are stored in the PostgreSQL database (path_config table), not hardcoded in the application. Changes made in the Path Config tab take effect immediately for the next scheduled or manual upload cycle.

Quick Reference – Error Summary

Error / Symptom	Probable Cause	Quick Fix
Last Execution Status: failed	Source path inactive or Azure unreachable	Check Path Config; verify Connectivity tab
Scheduler Status: Active, uploads not arriving	All paths are Inactive in Path Config	Activate at least one path in Path Config tab
Azure Blob Storage: Disconnected	Invalid connection string or rotated key	Update account key in azure_service.py
Logs show Response status: 401	Storage account key expired or invalid	Regenerate key in Azure Portal; update config
Logs show Response status: 403	Firewall or RBAC blocking storage access	Check Azure Storage firewall rules
Logs show Response status: 404	Container name wrong or does not exist	Confirm container 'spvdataflow' exists in storage account
No log entries appearing	Backend service not running or SDK error	Restart EL_BACKEND_AZURE service
UNC path not accessible	Network share offline or wrong credentials	Verify NETWORK_SHARE_PATH/USER/PASS in compose; remount share
Manual upload returns error	Source path not found or container unreachable	Confirm active path exists and Connectivity tab shows Connected